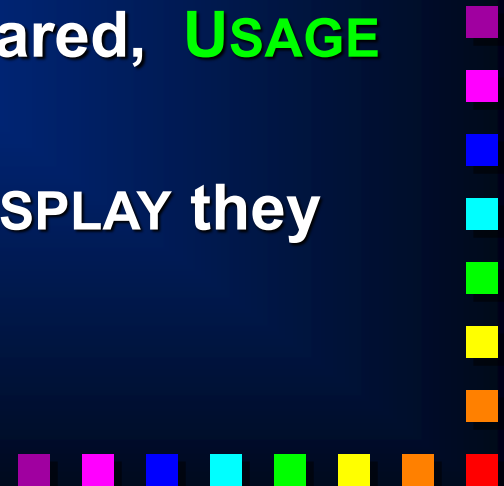


The USAGE clause

USAGE IS DISPLAY

SYSTEM	CHAR	HEX	DEC	8	4	2	1	8	4	2	1
ASCII	"A"	41	65	0	1	0	0	0	0	0	1
EBCDIC	"A"	C1	193	1	1	0	0	0	0	0	1

- ◆ The USAGE clause specifies the **format** of **numeric data items** in the computer storage.
- ◆ When no USAGE clause is explicitly declared, **USAGE IS DISPLAY** is assumed.
- ◆ When numeric items have a USAGE of DISPLAY they are held as **ASCII** characters !!



Arithmetic when USAGE IS DISPLAY

		8	4	2	1	8	4	2	1	HEX	CHAR
Num1	PIC 9 VALUE 4.	0	0	1	1	0	1	0	0	41	"4"
Num2	PIC 9 VALUE 1.	0	0	1	1	0	0	0	1	31	"1"
Num3	PIC 9.	0	0	0	0	0	0	0	0	0	nul

ADD Num1,Num2 GIVING Num3.

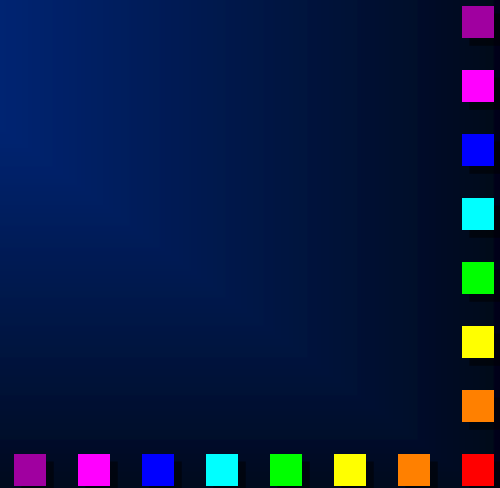
Arithmetic when USAGE IS DISPLAY

		8	4	2	1	8	4	2	1	HEX	CHAR
Num1	PIC 9 VALUE 4.	0	0	1	1	0	1	0	0	41	"4"
Num2	PIC 9 VALUE 1.	0	0	1	1	0	0	0	1	31	"1"
Num3	PIC 9.	0	1	1	0	0	1	0	1	65	"e"

ADD Num1,Num2 GIVING Num3.

Why use the USAGE clause?

- ◆ The USAGE clause is used for purposes of optimisation - both speed and storage.
- ◆ The USAGE clause controls the way data items (normally numeric) are stored in memory.



USAGE Syntax

[USAGE IS] { BINARY
COMPUTATIONAL
COMP
INDEX
PACKED – DECIMAL
DISPLAY }

01 Num1 PIC 9(5)V99 USAGE IS COMP.

01 Num2 PIC 99 USAGE IS PACKED-DECIMAL.

01 Idxltem USAGE IS INDEX.

01 GroupItems COMP.

02 Item1 PIC 999.

02 Item2 PIC 9(4)V99.

02 New1 PIC S9(5) COMP SYNC.



USAGE IS COMP

- ◆ COMP items are held in memory as pure binary two's complement numbers.

Number of Digits	Storage Required.
1 TO 4	1 WORD (2 Bytes)
5 TO 9	1 LONGWORD (4 Bytes)
10 TO 18	1 QUADWORD (8 Bytes)

01 TotalCount PIC 9(7) USAGE IS COMP.

Takes 4 bytes (1 LONGWORD) of storage.

COMP - Storage Requirements

2 Bytes can represent any 4 digit number.

32768	16384	8192	4096	2048	1024	512	256	128	64	32	16	8	4	2	1

**Max. number = + or - 32767
PIC 9999**

4 Bytes can represent any 9 digit number

**Max. number = + or - 2, 147,483,647
PIC 999 999 999**

USAGE IS PACKED-DECIMAL

- ◆ Data items declared as PACKED-DECIMAL are held in **binary-coded-decimal** (BCD) form.
- ◆ Instead of representing the value as one single binary number, each digit is held in binary form in a nibble (half a byte).

PIC S9
VALUE +5

5	+
---	---

PIC S9(2)
VALUE -32

0	3	2	-
---	---	---	---

PIC S9(3)
VALUE +262

2	6	2	+
---	---	---	---



The SYNCHRONIZED Clause

- ◆ The SYNCHRONIZED clause explicitly aligns COMP and INDEX data items along their natural WORD boundaries.
- ◆ If there is no SYNCHRONIZED clause the data items are aligned on byte boundaries.
- ◆ This clause is used to optimise speed of processing - but it does so at the expense of storage.

```
01 ThreeBytes  PIC XXX VALUE "DOG".  
01 TwoBytes    PIC 9(4)  COMP.
```



The SYNCHRONIZED Clause

COMP SYNC (1 TO 4 Digits)	= 2 Byte boundary
COMP SYNC (5 TO 9 Digits)	= 4 Byte boundary
COMP SYNC (10 TO 18 Digits)	= 8 Byte boundary
INDEX	= 4 Byte boundary

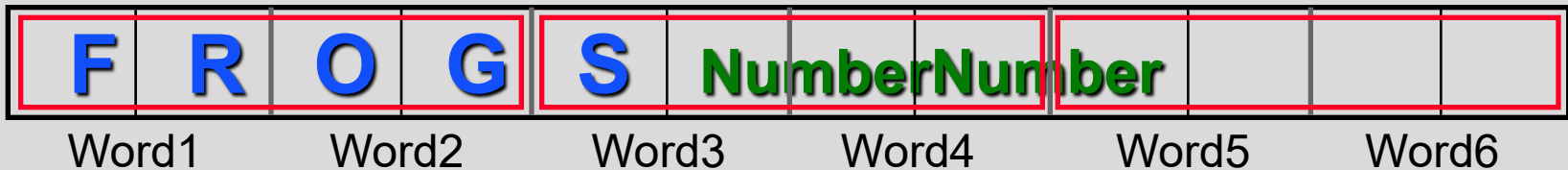
```
01 ThreeBytes  PIC XXX VALUE "DOG".  
01 TwoBytes    PIC 9(4)  COMP SYNC.
```



The SYNCHRONIZED Clause

COMP SYNC (1 TO 4 Digits)	= 2 Byte boundary
COMP SYNC (5 TO 9 Digits)	= 4 Byte boundary
COMP SYNC (10 TO 18 Digits)	= 8 Byte boundary
INDEX	= 4 Byte boundary

```
01 FiveBytes    PIC XXX VALUE "FROGS".  
01 FourBytes    PIC S9(5)  COMP.
```



The SYNCHRONIZED Clause

COMP SYNC (1 TO 4 Digits)	= 2 Byte boundary
COMP SYNC (5 TO 9 Digits)	= 4 Byte boundary
COMP SYNC (10 TO 18 Digits)	= 8 Byte boundary
INDEX	= 4 Byte boundary

01 FiveBytes PIC XXX VALUE "FROGS".
01 FourBytes PIC S9(5) COMP SYNC.

Aligns along a
4 byte boundary

